

Maven Lifecycle - Step by Step Detailed Guide

This guide explains the Maven lifecycle phases in a practical, step-by-step way.

1. Validate

Checks project structure, pom.xml correctness, and required files.

2. Compile

Compiles source code (src/main/java) into .class files in target/classes.

3. Test

Runs unit tests from src/test/java using frameworks like JUnit.

4. Package

Creates a distributable artifact such as JAR or WAR.

5. Verify

Runs integration tests and additional checks.

6. Install

Installs artifact into local repository (~/.m2/repository).

7. Deploy

Deploys artifact to remote repositories like Nexus or Artifactory.

Lifecycle Flow:

validate → compile → test → package → verify → install → deploy

Common Command:

```
mvn clean install
```

This command cleans old builds and runs the full lifecycle up to install.

Parent POM:

A Parent POM centralizes configuration and dependencies for multiple projects. Child projects inherit configurations, reducing duplication.

Dependency Scope:

Scope	Description
compile	Available in all phases and packaged
provided	Available at compile time, not packaged
runtime	Required at runtime only
test	Used only during testing
system	Local dependency (not recommended)